

Test Booklet

Series

A

Test Booklet No.

**Ph.D. Entrance Examination
COMPUTER ENGINEERING**

Name of Applicant Answer Sheet No.

Applicant ID/Roll No. : Signature of Applicant :

Date of Examination: Signature of the Invigilator(s)
1.

Time of Examination : 2.

Duration : 2 Hours]

[Maximum Marks : 100

IMPORTANT INSTRUCTIONS

- (i) The question paper is in the form of Test-Booklet containing **100 (Hundred)** questions. Each question carries four answers marked (A), (B), (C) and (D), out of which only one is correct.
- (ii) On receipt of the Test-Booklet (Question Paper), the candidate should immediately check it and ensure that it contains all the pages, i.e., **100** questions (50 in Part-I + 50 in Part-II). Discrepancy, if any, should be reported by the candidate to the invigilator immediately after receiving the Test-Booklet.
- (iii) A separate Answer-Sheet is provided with the Test-Booklet/Question Paper. On this sheet there are **100** rows containing four circles each. One row pertains to one question.
- (iv) The candidate should write his/her Application ID/Roll Number at the places provided on the cover page of the Test-Booklet/Question Paper and on the Answer-Sheet and NOWHERE ELSE.
- (v) No second Test-Booklet/Question Paper and Answer-Sheet will be given to a candidate. The candidates are advised to be careful in handling it and writing the answer on the Answer-Sheet.
- (vi) For every correct answer of the question **One (1) mark will be awarded**. For every unattempted question, Zero (0) mark shall be awarded. **There is no Negative Marking**.
- (vii) Marking shall be done only on the basis of answers responded on the Answer-Sheet.
- (viii) To mark the answer on the Answer-Sheet, candidate should **darken** the appropriate circle in the row of each question with Blue or Black pen.
- (ix) For each question only **one** circle should be **darkened** as a mark of the answer adopted by the candidate. If more than one circle for the question are found darkened or with one black circle any other circle carries any mark, the question will be treated as cancelled.
- (x) The candidates should not remove any paper from the Test-Booklet/Question Paper. Attempting to remove any paper shall be liable to be punished for use of unfair means.
- (xi) Rough work may be done on the blank space provided in the Test-Booklet/Question Paper only.
- (xii) *Mobile phones (even in Switch-off mode) and such other communication/programmable devices are not allowed inside the examination hall.*
- (xiii) No candidate shall be permitted to leave the examination hall before the expiry of the time.

DO NOT OPEN THIS QUESTION BOOKLET UNTIL ASKED TO DO SO.

Computer Engg.

[P.T.O.

PART-I
RESEARCH METHODOLOGY

1. The major responsibilities of a teacher are:
- I. To design curriculum and to ensure developmental goals and instructional objectives.
 - II. To design individualized learning plans for students, considering their unique needs and abilities.
 - III. To manage budgets, school policies and other events.
 - IV. To manage classroom dynamics and create a positive learning atmosphere.

Codes :

- (A) I, II, and III only.
- (B) I, III, and IV only.
- (C) I, II, and IV only.
- (D) All the four.

2. Which of the following doesn't characterize the nature of teaching?
- I. An interactive process
 - II. A specialized task
 - III. Subjected to analysis
 - IV. A system of actions

Codes :

- (A) III
- (B) Both II and III
- (C) Both III and IV
- (D) None of the four

3. Three statements are given followed by two conclusions numbered I and II. Assuming the statements to be true, even if they seem to be at variance with commonly known facts, decide which of the conclusions logically follow(s) from the statements?

Statements :

- 1. Some Pens are Pencils.
- 2. Some Pencils are Scales.
- 3. All Scales are Papers.

Conclusions :

- I. Some Scales are Pens.
 - II. Some Papers are Pencils.
- (A) Only conclusion I follows
 - (B) Only conclusion II follows
 - (C) Neither conclusion I nor II follows
 - (D) Both conclusions I and II follow

4. are the kind of things that learner encounters in his/her everyday life which helps them learn. For example, visiting an agriculture farm and see the crops closely for learning their pattern of growth more meaningfully.

- (A) Real-life aids
- (B) Interactive aids
- (C) Visual aids
- (D) Projected aids

5. Which of the following is not true about teaching?

- (A) Teaching is a set of events, outside the learners which are designed to support internal process of learning.
- (B) Teaching is a process in which learner, teacher, and curriculum are organized in a systematic way to attain educational goals.
- (C) The teaching is important because it is the only source of attaining learning, career excellence and success.
- (D) Teaching is an art and the art comes from the teacher's personality, experience, and talent.

6. The research that often takes the form of a field investigation and aims at collecting the basic data for verifying the applicability of existing theories and models in given situation is known as :
- Fundamental research
 - Theoretical research
 - Applied research
 - Conceptual research
7. Arrange the following steps of scientific research in a logical sequence.
- Formulating research problem
 - Extensive review of literature
 - Finalizing the variables
 - Testing the hypotheses
 - Collecting the data
 - Planning the research design
- Codes :**
- II-I-III-VI-V-IV
 - I-II -VI-III-V-IV
 - I-II -VI-III-IV-V
 - III-II -V-I-V-IV
8. A researcher intends to investigate differential efficacy of teaching methods in classroom learning. Hence, he draws three groups of students randomly and each of these groups was assigned one of the three teaching methods-lecture, discussion, and seminar at random. Most probably the researcher is impending to
- ex post facto research
 - experimental research
 - cross sectional research
 - survey research
9. Which of the following is not a step involved in the process of hypothesis testing?
- Deciding the level of significance.
 - Specifying type I or type II error.
 - Making a formal statement of hypothesis.
 - Choosing and applying a statistical test.
10. Which option will replace the question mark (?) in the following alpha-numeric series?
S 12, P 15, M 18, ? ?, G 24
- J 22
 - O 21
 - J 21
 - K 20
11. Which of the following points are the part of research ethics?
- carry out research in a competent manner
 - report results accurately
 - consider the consequences to society of any research endeavor
 - protect the rights and welfare of human participants.
 - consider the risk/benefit ratio for the proposed research
- Codes :**
- I, II, and IV
 - I, III, IV, and V
 - I, II, IV and V
 - All the five
12. If "A @ B" means A is husband of B; "A # B" means A is son of B; "A % B" means A is sister of B, then how is P related to M if M # J @ L % P, while P is a male?
- Paternal uncle
 - Maternal uncle
 - Uncle-in-law
 - Cousin

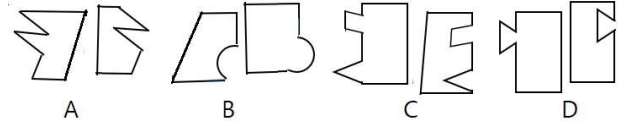
13. Which of the following may serve as barriers in the way of communication.

I. Pre-conceived attitudes
II. Cultural factors
III. Language
IV. Inconsistency
V. Perception

Codes :

- (A) II, III, and V
(B) I, II, III, and V
(C) II, III, IV, and V
(D) I, II, III, IV, and V
14. An academic event being organized in a sufficiently large room or a lecture hall with all participants attending the entire program in a single day. Besides regular presentations, the event includes sessions that invite more discussion and exchange of ideas, such as panel discussions. The presenters and panelists are renowned experts in the particular area of the discourse. Such an event is called
- (A) Workshop
(B) Seminar
(C) Symposium
(D) Conference
15. In a certain code language if ALMIRAH is written as MLAIHAR, how will CARPENTER be written in the same code language?
- (A) PRACRETNE
(B) PRACERETN
(C) RACNEPRET
(D) RETNECARP

16. There of the following four figures are alike in a certain way and one is different. Select the odd one.



(A) B
(B) D
(C) A
(D) C

17. Which of the following guidelines necessarily be followed while preparing general format of research paper as per the APA style?

I. Paper should be typed and 1.5 spaced on standard-sized paper.
II. The margins of the paper should be set to 1 inch on all sides.
III. There should also only be one space after punctuation marks such as a period.
IV. Paper should include a running manuscript head in the upper right-hand corner of each page.
V. The pages of the paper should be numbered consecutively, beginning with the title page in the upper right corner of each page.

Codes :

(A) I, II, III, and IV
(B) I, III, IV, and V
(C) I, II, III and V
(D) All the five

Read the following passage and answer the Question Nos. 18 to 22 :

A great industrialist Mr. Joseph told me one time how he started in a small store. He said that he used to dream of a large corporation with branches all over the country. He added that regularly and systematically he pictured in his mind the giant building, offices, factories, and stores, knowing that through the transformation of the mind, he could weave the fabric out of which his dreams would be clothed. He prospered, and began to attract to himself—by the universal law of attraction—the ideas, personnel, friends, money, and everything needed for the unfoldment of his ideal. He truly exercised and cultivated his imagination, and lived with these mental patterns in his mind until imagination clothed them in form.

Joseph believed, "It is just as easy to imagine yourself successful as it is to imagine failure, and far more interesting." He is a dreamer, and a dreamer of dreams. This means he has visions, images, and ideals in his mind, and knows that there is a creative power that responds to his mental pictures. The mental images we hold are developed in feeling. It is wisely said that all our senses are modifications of the one-sense-feeling. So, he dreamed a dream, and told it to his brethren, and they hated him. To hate is to reject in a simple language. The thoughts, feelings, beliefs, and opinions in your mind are the brethren that challenge you, belittle your dreams, and say to you, "You can't, it is impossible, forget it!"

Perhaps other thoughts come into your mind that ridicule your plan or ambition. You discover that there is a quarrel in your mind with your own brethren; opposition sets in. The way to handle the opposition in your mind is to detach

your attention from all potential distractors and begin to think clearly and with interest about your goal or objective. When your mind is engaged on your goal or objective, you are using the creative law of mind, and it will come to pass.

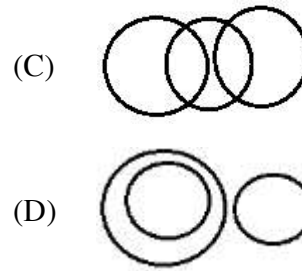
18. What Mr. Joseph refers to the term brethren?
 - (A) his colleagues
 - (B) his brothers
 - (C) his own psyche
 - (D) the other persons
19. Which of the following could be the most prudent title of the passage?
 - (A) Dreams come true
 - (B) Believe in yourself
 - (C) Power of feelings
 - (D) Meet the challenge
20. According to the passage, how can we manage opposing thoughts and ideas?
 - (A) Igniting creative power that responds to mental pictures, thoughts, and feelings.
 - (B) Have a dream, a plan, or purpose that you would like to accomplish.
 - (C) Detach your attention from distractors and focus keenly on your goal or objective.
 - (D) Settle the quarrel in your mind with your own brethren.
21. What tends to ridicule your plan or ambition?
 - (A) Pre-conceived thoughts and beliefs
 - (B) People those hate your idea
 - (C) Your own brethren
 - (D) The thought, it is impossible

22. What Mr. Joseph means by the universal law of attraction?
- Attracting to mental patterns in the mind
 - Attraction to his own mental pictures and dreams
 - Attraction to all resources needed for the goal/dream
 - Attraction to valued things-friends, money etc.

23. If there is a discrepancy between the message conveyed verbally and non-verbally, then
- verbal message is believed to be trustworthy
 - non-verbal message is believed to be trustworthy
 - none of them can be trustworthy
 - both of them have their own relevance

24. Which of the following is not amongst the seven basic principles (abbreviated as 7 Cs) of communication skills?
- Concise
 - Common
 - Courteous
 - Correct

25. Select the Venn diagram that best illustrates the relationship between the following classes :
Broccoli, Guava, Vegetables



26. In the process of communication, the step which converts thoughts or ideas into a communicable format is called

- Decoding
- Sender's input
- Transmission
- None of these

27. The biggest award for making measurable and major impact in the protection of environment in India is constituted in the name of

- Sunderlal Bahuguna
- Indira Gandhi
- Lal Bahadur Shastri
- Atal Bihari Vajpayee

28. Which of the following are the primary goals of communication?

- It monitors the performances of the employees at all the levels.
- It establishes and informs the members of the organisation about its goals.
- It helps arranging the finances and making budgetary provisions.
- It helps developing plans for the achievement of organizational goals.
- It encourages suggestions from subordinates in work conditions.

Codes :

- I, II, IV, and V
- I, III, IV, and V
- I, II, III, and V
- All the five

29. The phenomena of accumulation of non-biodegradable pesticides in human beings is called :
- (A) Biodegradation
 - (B) Bioremediation
 - (C) Biomagnification
 - (D) Bioaccumulation
30. LIFE : 10743 :: PICK : ?
- (A) 158210
 - (B) 1710412
 - (C) 14719
 - (D) 14829
31. Which of the following factors can make the communication process ineffective?
- I. Information overload
 - II. Assumptions and stereotyping
 - III. Emotional barriers
 - IV. Non-Verbal inconsistency
 - V. Inadequate feedback
- Codes :**
- (A) I, II, IV, and V
 - (B) I, III, IV, and V
 - (C) II, III, IV, and V
 - (D) I, II, III, and IV
32. Three of the following four letter-cluster pairs are alike in a certain way and one is different. Pick the odd one.
- (A) DWFU-HSJQ
 - (B) HSJQ-NMPK
 - (C) BYDW-GTIR
 - (D) MNOL-SHUF
33. Which of the following organizations looks after the quality of technical and management education in India?
- (A) AICTE
 - (B) CSIR
 - (C) MCI
 - (D) NCTE
34. Which of the following is true in respect of primary and secondary data?
- I. Primary data are cost efficient
 - II. Secondary data are more authentic
 - III. Secondary data are rated high in time economy
 - IV. Primary data have the ease of analysis
- Codes :**
- (A) III and IV
 - (B) I and III
 - (C) II and IV
 - (D) I, II, and III
35. Mr. Anthony, an electrical engineer desired to analyze the load requirement in peak winter and peak summer in 'Y' locality. Apart from other predictions, he hypothesized that "the load requirement in summer is significantly higher than winter." He basically surveyed actual electricity consumption during three months each of peak summer and peak winter on hourly basis. The data analyzed through t-test revealed significant difference ($p < .01$) between the mean value of electricity consumption during summer and winter

- season. What inference would you drawn in respect of the proposed hypothesis?
- (A) The hypothesis is accepted
 - (B) The hypothesis is rejected
 - (C) Information is deficient
 - (D) Analysis applied is not fit for such hypothesis
36. The benefits of ICT over conventional communication system are :
- I. Alternate to physical teacher
 - II. Richer educational resources
 - III. Cost effective
 - IV. Richer and quality education
 - V. Multitude of communication
- Codes :**
- (A) I, II, III, and V
 - (B) II, III, and V
 - (C) II, III, IV, and V
 - (D) All the above
37. Which of the following is not true about quantitative data in respect of primary and secondary data?
- I. Primary data are cost efficient
 - II. Secondary data are more authentic
 - III. Secondary data are rated high in time economy
 - IV. Primary data have the ease of analysis
- Codes :**
- (A) III and IV
 - (B) I and III
 - (C) II and IV
 - (D) I, II, and III
38. Which of the following is essential component of communication cycle?
- (A) Internet connection
 - (B) An e-mail ID
 - (C) An interpreter
 - (D) A message
39. In mega cities of India, the major source of air pollution is :
- (A) Thermal power
 - (B) Transport sector
 - (C) Municipal waste
 - (D) Commercial sector
40. A researcher collected primary (quantitative) data from 8th (N = 180) and 12th (N = 184) grade students by employing a scale of social confidence. All the participants were tested for social confidence on three occasions, i.e., at initial stage of the study, at the end of one-week social confidence building training, and after one month of training (follow up). What analysis do you suggest for main and interactive effects of grade level and training?
- (A) Multiple comparison test
 - (B) 2×3 way ANOVA
 - (C) 2×3 way ANOVA (independent means)
 - (D) 2×3 way ANOVA (repeated measures)
41. What does PPTP stands for?
- (A) Point to Point Tunneling Protocol
 - (B) Point to Point Transfer Protocol
 - (C) Point to Point Traffic Protocol
 - (D) Point to Point Transmission Protocol

42. Which of the following does not characterize qualitative data?
- Helpful for summarizing large amounts of information in a snapshot.
 - Provide rich description and context about a complicated issue.
 - Amenable to coding, editing and objective analysis.
 - Generally based on ideographic approach
- Codes :**
- III
 - I and III
 - I and IV
 - I, III, and IV
43. The most recurring natural hazard in India is :
- Landslides
 - Earthquakes
 - Cloudbursts
 - Floods
44. A business scientist is studying sale pattern of six top trending car models during past 12 months. What could be the most pertinent graphical presentation of his data so as to make understandable for the prospective buyers?
- Line graph
 - Bar diagram
 - Pie graph
 - Scatter graph
45. Which part of the CPU performs calculations and makes decisions?
- Calculation logic unit
 - Alternating logic unit
 - Arithmetic logic unit
 - Alternate local unit
46. Who is vested with the Chancellorship of the Government State Universities in India?
- Any person appointed by CM
 - Education minister of the state
 - Any person appointed by Governor
 - Governor of the state
47. Which one of the following is not a network device?
- Hub
 - Switch
 - Bridge
 - Router
48. Which of the following tool is most likely to offer objective and reliable data?
- Structured interview
 - Case history
 - Systematic observation
 - Rating scale
49. When was the first e-mail successfully launched?
- 1969
 - 1971
 - 1966
 - 1973
50. Indian Institute of Advance Study is located at :
- Delhi
 - Chandigarh
 - Shimla
 - Bangalore

PART-II
COMPUTER ENGINEERING

51. Cauchy's Integral Theorem is related to _____.
(A) Line integrals
(B) Surface integrals
(C) Contour integrals
(D) Volume integrals
52. What method is commonly used for solving systems of linear algebraic equations?
(A) Newton-Raphson method
(B) Euler's method
(C) Gaussian elimination
(D) Runge-Kutta method
53. In a singular square matrix, the determinant is _____.
(A) Non-zero
(B) Equal to zero
(C) Infinite
(D) Undefined
54. If A is a square matrix and where A is invertible, then the solution to $Ax = b$, is unique if and only if _____.
(A) A is singular
(B) $\det(A) = 0$
(C) $\det(A) \neq 0$
(D) Matrix b has all zeros
55. Rolle's Theorem states that if $f(a) = f(b)$ and $f(x)$ is continuous on $[a, b]$, then there exists a c in (a, b) such that _____.
(A) $f'(c) = 0$
(B) $f''(c) = 0$
(C) The integral of $f(x)$ from a to b is 0
(D) $f(c)$ is the maximum value of $f(x)$ on $[a, b]$
56. Laplace transform converts ordinary differential equations into _____.
(A) Algebraic equations
(B) Integral equations
(C) Difference equations
(D) None of the above
57. In a room containing 28 females, there are 18 females who can speak English, 15 females speak French and 22 speak German. 9 females speak both English and French, 11 females speak both French and German whereas 13 speak both German and English. How many females speak all the three languages?
(A) 9
(B) 8
(C) 6
(D) 7

58. Which algorithm efficiently computes the convex hull of a set of points in a plane?
- (A) Kruskal's Algorithm
 - (B) Prim's Algorithm
 - (C) Prim's Algorithm for Minimum Spanning Tree
 - (D) Graham Scan Algorithm
59. What is the main advantage of using a self-balancing search tree like an AVL tree over a standard binary search tree?
- (A) Increased storage efficiency
 - (B) Faster element insertion
 - (C) Guaranteed logarithmic worst-case search time
 - (D) Simpler implementation
60. Which algorithm paradigm guarantees optimal solutions for problems with well-defined subproblems and optimal solutions for those subproblems?
- (A) Greedy Algorithms
 - (B) Dynamic Programming
 - (C) Divide-and-Conquer
 - (D) Brute Force
61. What is the main advantage of using a good hash function in hash tables?
- (A) Faster retrieval
 - (B) Lower space complexity
 - (C) Avoidance of collisions
 - (D) Reduced insertion time
62. The KMP algorithm is used for efficient _____.
- (A) Sorting
 - (B) Searching
 - (C) Pattern matching
 - (D) None of the above
63. The following numbers are inserted into an empty binary search tree in the given order : 101, 520, 1100, 82, 79, 21, 65, 721, 855, 919, 19. What is the height of the binary search tree ($H = L$; $L = 0$), number of nodes in left subtree, and number of nodes in right subtree?
- (A) 5, 5, 5
 - (B) 5, 6, 4
 - (C) 6, 7, 3
 - (D) 6, 5, 5
64. Which data structure efficiently performs membership queries with an average time complexity of $O(1)$?
- (A) Binary search tree
 - (B) Linked list
 - (C) Hash table
 - (D) Array

65. Consider the following recurrence:
 $T(n) = 2 * T(\text{ceil}(\sqrt{n})) + 1, T(1) = 1.$
 Which one of the following is true?
- (A) $T(n) = \Theta(\log \log n)$
 - (B) $T(n) = \Theta(\log n)$
 - (C) $T(n) = \Theta(\sqrt{n})$
 - (D) $T(n) = \Theta(n)$
66. What type of finite automaton has an output that depends only on the current state?
- (A) Mealy machine
 - (B) Moore machine
 - (C) Hybrid FSM
 - (D) Encapsulated FSM
67. Which problem is proven to be undecidable in the context of Turing machines?
- (A) Halting problem
 - (B) Sorting problem
 - (C) Shortest path problem
 - (D) Primality testing
68. What type of language can be accepted by a Turing machine that may not halt on every input?
- (A) Regular language
 - (B) Context-free language
 - (C) Recursively enumerable language
 - (D) Context-sensitive language
69. Which of the following is the correct statement?
- (A) A DFA has always same number of states as compared to an equivalent NFA.
 - (B) A DFA has less number of states as compared to an equivalent NFA.
 - (C) A DFA has same or more number of states as compared to an equivalent NFA.
 - (D) A DFA has more expressive power as compared to NFA.
70. Consider the virtual page reference string 8, 2, 3, 2, 8, 5, 4, 8, 3, 5, 2, 4, 8. On a demand paged virtual memory system running on a computer system that main memory size of 4-page frames which are initially empty. Let LRU, FIFO and OPTIMAL denote the number of page faults under the corresponding page replacements policy. Then
- (A) LRU = 9, FIFO = 7, OPTIMAL = 6
 - (B) LRU = 9, FIFO = 7, OPTIMAL = 7
 - (C) LRU = 10, FIFO = 7, OPTIMAL = 6
 - (D) LRU = 9, FIFO = 8, OPTIMAL = 6
71. Which scheduling algorithm is designed to minimize the waiting time of processes?
- (A) First-Come-First-Serve (FCFS)
 - (B) Shortest Job Next (SJN)
 - (C) Round Robin (RR)
 - (D) Priority Scheduling

72. Match List -I with List-II :

List-I

- a. Inconsistency
- b. Deadlock resolution
- c. Deadlock Avoidance
- d. Transit

List-II

- 1. Require the recognition of unsafe states
- 2. A message received by P_j from P_i that is not shown as sent by P_i
- 3. A message sent by P_i to $P(m_{ij})$ that does not belong to the set of message received by P_j
- 4. Always requires the abortion of one or more executing processes

Select the most appropriate answer from the options given below :

- (A) a-2 b-3 c-4 d-1
- (B) a-4 b-2 c-3 d-1
- (C) a-2 b-4 c-1 d-3
- (D) a-1 b-3 c-2 d-4

73. Threads are lightweight processes that share what resources with their parent process?

- (A) CPU time
- (B) Memory address space
- (C) File handles
- (D) All of the above

74. Given relations $p(c, d)$ and $q(e, f)$, the result of select distinct c, d from p, q is guaranteed to be same as p , provided

- (A) p and q have the same number of tuples
- (B) p and q have no duplicates
- (C) q has no duplicates and p is non-empty
- (D) p has no duplicates and q is non-empty

75. Match the following List-I with List-II :

List-I

- a. Data security
- b. Foreign keys
- c. Private key
- d. Event control action model

List-II

- (i) Trigger
- (ii) Domain constraint
- (iii) Encryption
- (iv) Referential integrity

Select the correct answer from the options given below:

- (A) a-(iii) b-(i) c-(iv) d-(ii)
- (B) a-(iii) b-(iv) c-(ii) d-(i)
- (C) a-(iv) b-(ii) c-(iii) d-(i)
- (D) a-(ii) b-(i) c-(iii) d-(iv)

76. Aggregate functions in SQL are _____.

- (A) Large, Least & Absolute
- (B) Upper, Sum & Greatest
- (C) Sqrt, Count, Avg
- (D) Sum, Avg & Max

77. What type of data does a spatial database primarily deal with?
- (A) Textual data
 - (B) Numeric data
 - (C) Geographical data
 - (D) Multimedia data
78. For a database relation R (a, b, c, d) where the domains of a, b, c and d include only atomic values. The functional dependency $a \rightarrow c, b \rightarrow d$ holds in the relation is in
- (A) In 2NF not in 3NF
 - (B) In 1NF not in 2NF
 - (C) In BCNF
 - (D) In 3NF
79. A virtual memory system with 32 bit virtual addresses uses 4 KB page size and 128 MB byte-addressable main memory. What is the virtual page number size (in bit) for this memory system?
- (A) 20
 - (B) 15
 - (C) 32
 - (D) 27
80. Which of the following is not a hindrance factor to instruction pipeline processing?
- (A) Data Dependencies
 - (B) Resource Dependencies
 - (C) Transitive Dependencies
 - (D) Control Dependencies
81. Which addressing mode allows operand encoded in the instruction itself?
- (A) Register addressing
 - (B) Immediate addressing
 - (C) Direct addressing
 - (D) Indirect addressing
82. Which of the following is not the reason of cache coherence problem in shared memory MIMD Processors?
- (A) Sharing of writable data
 - (B) Process Migration
 - (C) I/O activity
 - (D) Multithreading
83. Arrange the following operators used in Boolean expressions in the order of their precedence from highest to lowest.
- (i) OR
 - (ii) Parenthesis
 - (iii) NOT
 - (iv) AND
- Choose the correct answer from options given below :
- (A) (iii), (iv), (i), (ii)
 - (B) (ii), (i), (iv), (iii)
 - (C) (ii), (iii), (iv), (i)
 - (D) (ii), (i), (iii), (iv)

84. Which type of parallel system shares a single memory space accessible by all processors?
- (A) Multi-computer
 - (B) Multiprocessor
 - (C) Array processor
 - (D) Vector processor
85. Grid computing leverages geographically distributed resources to _____.
 (A) Increase computational power
 (B) Improve availability
 (C) Enhance fault tolerance
 (D) All of the above
86. Choose invalid statement in context of software testing:
- (A) System testing is usually done by the development team.
 - (B) Release testing is performed after acceptance testing
 - (C) Release testing is usually a black box testing.
 - (D) Alpha testing is a type of user testing.
87. In Aspect-Oriented Programming (AOP), what is an "Aspect"?
- (A) A function or method
 - (B) A programming language
 - (C) A cross-cutting concern
 - (D) A design pattern
88. Arrange the phases of requirement analysis in their proper order.
- (i) Develop prototypes
 - (ii) Draw the context diagrams
 - (iii) Model the requirements
 - (iv) Finalize the requirements
- (A) (i), (ii), (iii), (iv)
 - (B) (ii), (i), (iii), (iv)
 - (C) (ii), (i), (iv), (iii)
 - (D) (i), (iii), (iv), (ii)
89. Match the following List-I with List-II :
- List-I**
- a. Mean Time To Failure
 - b. Path coverage adequacy criteria
 - c. McCabe's Cyclomatic Complexity
 - d. Function Point
- List-II**
- 1. Measure of the number of linearly independent paths through a program's source code.
 - 2. Software Reliability Metric
 - 3. Unit of measurement to express the amount of business functionality
 - 4. White Box Testing
- Select the correct answer from the options given below :
- (A) a-2 b-3 c-4 d-1
 - (B) a-4 b-1 c-2 d-3
 - (C) a-3 b-2 c-4 d-1
 - (D) a-2 b-4 c-1 d-3

90. Petri nets are useful for _____.
 (A) Representing concurrent and asynchronous systems
 (B) Designing user interfaces and menus
 (C) Implementing database schemas
 (D) Writing unit tests
91. Determine the maximum length of the cable (in km) for transmitting data at a rate of 600 Mbps in an Ethernet LAN with frames of size 10,000 bits. Assume the signal speed in the cable to be 3,00,000 km/s.
 (A) 2
 (B) 2.5
 (C) 3
 (D) 5
92. With respect to the channel access methods in broadcast network, which of the following statements is incorrect?
 (A) CSMA/CD is a fairer method than Token Passing.
 (B) Token Passing is a fairer method than CSMA/CD.
 (C) Token Passing is more stable in heavy traffic conditions.
 (D) Adding and removing nodes to CSMA/CD LAN is easier than to Token Passing LAN.
93. Socket address is _____.
 (A) IPv4 address translated to IPv6 address
 (B) Combination of IP address and port address
 (C) Combination of MAC address and domain address
 (D) Application layer address
94. Match the following List-I with List-II:
List-I
 a. TELNET
 b. FTP
 c. NNTP
 d. DNS
List-II
 (i) Mapping host names on to their network address
 (ii) Move data efficiently from one machine to another
 (iii) Allows user on one machine to log into a distant machine and work there
 (iv) Moving new articles around
- Select the correct answer from the options given below:
 (A) a-(iii) b-(i) c-(iv) d-(ii)
 (B) a-(iii) b-(ii) c-(iv) d-(i)
 (C) a-(iv) b-(ii) c-(iii) d-(i)
 (D) a-(iii) b-(ii) c-(i) d-(iv)

95. Lexical analysis breaks down the source code into what basic units?
- (A) Lines of code
 - (B) Individual characters
 - (C) Tokens (keywords, identifiers, operators, etc.)
 - (D) Machine instructions
96. Parsers analyze the structure of the program according to which?
- (A) Regular expressions
 - (B) Context-free grammars
 - (C) Machine code instructions
 - (D) Flowcharts
97. Code optimization aims to improve _____ aspects of the generated code.
- (A) Readability and maintainability
 - (B) Runtime speed and efficiency
 - (C) Memory usage
 - (D) All of the above
98. The Flood Fill algorithm _____.
- (A) May use 4-connected or 8-connected approach
 - (B) Must use 8-connected approach
 - (C) Must use 4-connected approach
 - (D) Must not use recursive approach
99. The amount of light emitted by the phosphor coating depends on the?
- (A) Number of electrons striking the screen
 - (B) Speed of electrons striking the screen
 - (C) Distance from the cathode to the screen
 - (D) None of the above
100. Web multimedia technologies like audio and video are typically embedded using _____.
- (A) JavaScript
 - (B) HTML5
 - (C) CSS
 - (D) XML

ROUGH WORK

ROUGH WORK